LANCE

An autonomous and verifiable auditor for IoT/IIoT networks

23 October 2026 · Université de Mons, ILIA

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The objective

Under which conditions does an agentic approach prevail over a scripted one, rather than which model prevails?

- An auditor that explores, identifies, verifies and reports.
- Identical tools and budget for both approaches, within the control harness.
- Evidence recomputed from the log, not asserted.

Model training is not a current priority.

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The platform in brief

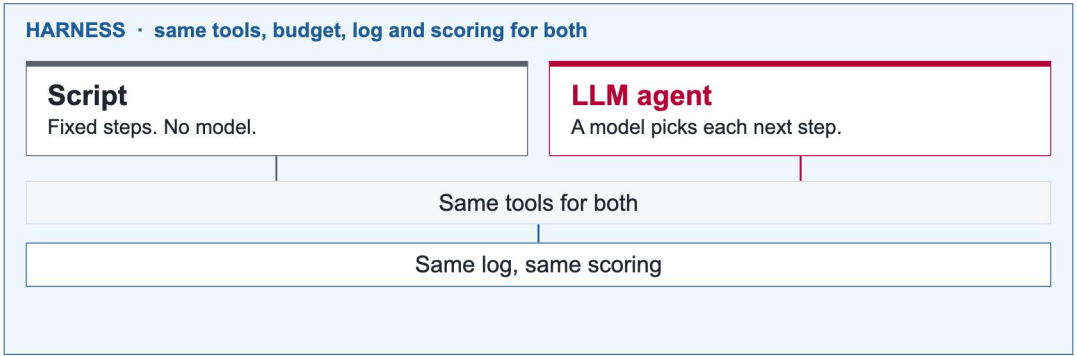
- **Runs the audit in six phases**
- **Four execution modes**
- **Every run kept as evidence**



The platform runs the audit; the benchmark judges it.

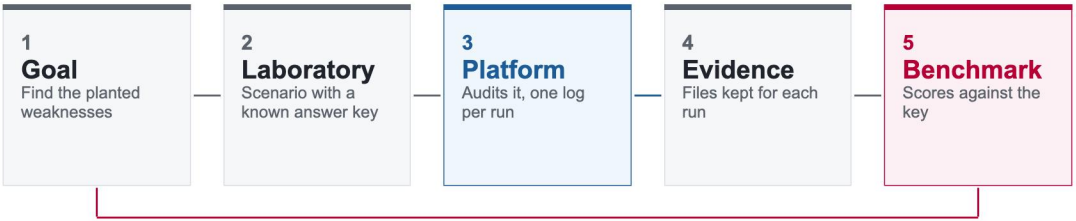
Captures of the running instance, 22 October 2026.

Script, LLM agent and harness



Only the step-choice mechanism changes.

How a run is judged against the goal



Each run is scored against the known weaknesses of its scenario.

What constitutes a finding

Healthy network

✕ Error

Any finding reported here is an error.

Vulnerable network

✓ Finding

Recomputed from the log before it counts.

The finding is recomputed from the log; the conclusion is never taken on trust.

A control task before testing the model

V0 is a control task, simple enough to be solved without inference. Both approaches must pass it.

1	2	Gate
Script passes	**Model passes**	**Both pass: the hypotheses may be read.**

Only a control that both approaches pass permits the hypotheses to be read.

Calibration passed on the lab network (3 of 3 runs per arm). The held-out instance remains outstanding.

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The objective restated as hypotheses

The register is this list: each hypothesis, with the test that decides it.

H1	**Connected devices**	a link absent from public inputs
H2	**Heterogeneous formats**	a representation no parser covers
H3	**Novel contracts**	rights stated in unseen documentation
H4	**Vendor semantics**	a field whose meaning is only documented
H5	**Ambiguous refusals**	resuming the audit after a refusal
H6	**Hostile content**	an instruction obeyed from observed content

H1–H5 quantify a possible gain. H6 quantifies a risk.

Next steps

1. Execute V0 on a held-out instance
2. Develop a parameterisable test harness
3. Build the scenario families for H2, H3 and H4
4. Deploy a reproducible platform

Support requested

- Access to real vendor artifacts
- An adversarial review of the register

Order follows conditions, not dates.

AsiaCCS 2027 submission deadline: 11 December 2026, anywhere on Earth.

Reserve — declared coverage gaps

- Physical impact and functional safety
- Device fragility under probing
- Real fleet scale
- Proprietary industrial protocols that are not simulated
- Complete multi-step exploitation chains
- Abstention and calibration
- Generic IT scenarios, deliberately excluded

A benchmark that states its gaps is stronger than one that claims full coverage.